

# Solution to Ramanujan Challenge Problem 3.1: An integral over knot polynomial roots expressing $\pi^2$

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## Abstract

Let  $A_{7_2}(M, L)$  be the  $A$ -polynomial of the knot  $7_2$ , let  $\alpha \approx 0.349269$  and  $\beta \approx 0.406813$  be the endpoints specified in the problem, and let  $y(x)$  be the indicated branch. We prove

$$\int_{\alpha}^{\beta} \left( \log x \frac{dy}{y} - \log y \frac{dx}{x} \right) = \frac{4\pi^2}{85}.$$

By Khoi's variation formula the integral is a difference of Godbillon–Vey invariants of the endpoint representations; the  $\beta$  endpoint is the maximal Fuchsian representation of  $\Sigma(2, 3, 17)$  and is evaluated by Brooks–Goldman. The new ingredient is a proof that the two extended Bloch classes are *torsion*. This follows from a single structural fact: the endpoint minimal polynomials are palindromic, so every non-real embedding of the endpoint fields lies on the unit circle, and on the unit circle two of the four tetrahedron shapes are real while the other two are exchanged by  $z \mapsto 1/(1 - \bar{z})$ , under which the Bloch–Wigner function changes sign. Torsion gives rationality; the Merkurjev–Suslin torsion order supplies the denominator bound 2040; a 301-digit evaluation then determines the value uniquely.

## 1 Setup

Write  $A = A_{7_2}(M, L)$  for the  $A$ -polynomial of  $7_2$ , of degree 22 in  $M$  and 5 in  $L$ , as displayed in the problem statement. Put

$$\alpha : A(\alpha, \alpha^{1/2}) = 0, \quad \beta : A(\beta, \beta) = 0,$$

the roots nearest 0.349269 and 0.406813, and let  $y(x)$  be the branch with  $y > 0$ ,  $y' \leq -2$  on  $[\alpha, \beta]$ . Set

$$I := \int_{\alpha}^{\beta} \left( \log x \frac{dy}{y} - \log y \frac{dx}{x} \right).$$

It is convenient to use the eigenvalue coordinate. Writing  $M = a^2$ ,  $L = a$  at the  $\alpha$  endpoint and  $M = L = b$  at the  $\beta$  endpoint, and factoring  $A(a^2, a)$  resp.  $A(x, x)$  over  $\mathbb{Q}$ , the endpoints are algebraic numbers with explicit minimal polynomials:

**Proposition 1** (Endpoint fields). *The endpoint eigenvalues satisfy*

$$f_{\alpha}(a) = a^{12} - 3a^{11} + 4a^{10} - 5a^9 + 6a^8 - 7a^7 + 7a^6 - 7a^5 + 6a^4 - 5a^3 + 4a^2 - 3a + 1, \quad (1)$$

$$\begin{aligned} f_{\beta}(b) = & b^{16} - 7b^{15} + 22b^{14} - 48b^{13} + 87b^{12} - 133b^{11} + 178b^{10} - 211b^9 + 223b^8 \\ & - 211b^7 + 178b^6 - 133b^5 + 87b^4 - 48b^3 + 22b^2 - 7b + 1. \end{aligned} \quad (2)$$

Both are irreducible over  $\mathbb{Q}$  and palindromic. Their signatures are  $(r_1, r_2) = (2, 5)$  and  $(4, 6)$ , and

$$a = 0.59098942867025644 \dots, \quad a^2 = \alpha, \quad b = 0.40681308133679 \dots = \beta.$$

## 2 Reduction to a regulator difference

Khoi's variation formula for the one-form  $\log x d \log y - \log y d \log x$  on an  $A$ -polynomial curve identifies  $I$  with a difference of Godbillon–Vey invariants of the two endpoint representations of  $\pi_1$  of the  $7_2$  complement, with the normalisation

$$I = \frac{\text{GV}(\rho_\alpha) - \text{GV}(\rho_\beta)}{4}. \quad (3)$$

**Proposition 2** ( $\beta$  endpoint).  $S_{-1}^3(7_2) \cong \Sigma(2, 3, 17)$ , the  $\beta$  representation factors through  $\pi_1(\Sigma(2, 3, 17))$  as the maximal Fuchsian representation, and

$$\text{GV}(\rho_\beta) = \frac{242\pi^2}{51}.$$

This is Brooks–Goldman applied to the Seifert data of  $\Sigma(2, 3, 17)$ : with  $\chi_{\text{orb}} = -11/102$  and  $e = -1/102$ , the maximal value is  $4\pi^2\chi_{\text{orb}}^2/e = -242\pi^2/51$  up to orientation.

By (3) and Proposition 2, the conjecture  $I = 4\pi^2/85$  is *equivalent* to

$$\text{GV}(\rho_\alpha) = \frac{242\pi^2}{51} + \frac{16\pi^2}{85} = \frac{1258\pi^2}{255} = \frac{74\pi^2}{15}. \quad (4)$$

Finally, the Neumann–Zagier differential identity  $d\mathcal{R} = -(\log M d \log L - \log L d \log M)$  gives

$$I = -\Delta\mathcal{R}, \quad \mathcal{R} = \sum_{j \in \{T, U, V, W\}} \widehat{R}(z_j), \quad \widehat{R}(z) = \text{Li}_2(z) + \frac{1}{2} \log z \log(1 - z) - \frac{\pi^2}{6}, \quad (5)$$

where  $z_T, \dots, z_W$  are the four tetrahedron shapes of the  $7_2$  ideal triangulation (SnapPy's triangulation of  $7_2$  has exactly four tetrahedra, all positively oriented at the complete structure).

## 3 The deformation chart

With  $X = M^2$  put

$$u = \frac{L + X^3}{X(L + X)}, \quad r = \frac{-(1 + \sqrt{1 + 4u^2})}{2u}, \quad T = 1 - r^2, \quad U = u, \quad V = \frac{u}{X}, \quad W = \frac{1}{1 - uX}. \quad (6)$$

At the two endpoints this gives, to the displayed precision,

$$\begin{aligned} T_\alpha &= -0.15787528\dots, & U_\alpha &= 6.8157987\dots, & V_\alpha &= 55.872474\dots, & W_\alpha &= 5.9329217\dots, \\ T_\beta &= -0.25828783\dots, & U_\beta &= 4.3429623\dots, & V_\beta &= 26.241958\dots, & W_\beta &= 3.5555141\dots, \end{aligned}$$

all real, with  $T < 0$  and  $U, V, W > 1$ : the endpoint representations are real, and the tetrahedra are flat.

The chart (6) involves  $\sqrt{1 + 4u^2}$ , so a priori the shapes lie in a quadratic extension of the endpoint field. They do not.

**Lemma 3** (Shape field).  $1 + 4u^2$  is a square in the endpoint field itself. Explicitly, with  $s_\alpha, s_\beta \in \mathbb{Z}[z]$

$$\begin{aligned} s_\alpha &= 12z^{11} - 30z^{10} + 32z^9 - 42z^8 + 50z^7 - 58z^6 + 54z^5 - 56z^4 + 42z^3 - 38z^2 + 28z - 21, \\ s_\beta &= 60z^{15} - 394z^{14} + 1150z^{13} - 2386z^{12} + 4196z^{11} - 6176z^{10} + 8022z^9 - 9202z^8 \\ &\quad + 9406z^7 - 8592z^6 + 6958z^5 - 4960z^4 + 3064z^3 - 1546z^2 + 646z - 137, \end{aligned}$$

one has  $s_\alpha(a)^2 = 1 + 4u^2$  in  $\mathbb{Q}(a)$  and  $s_\beta(b)^2 = 1 + 4u^2$  in  $\mathbb{Q}(b)$ . Hence all four shapes lie in the endpoint field, and the shape field  $E$  coincides with  $F$ .

*Proof.* Writing  $u$  itself as a polynomial modulo the minimal polynomial,

$$\begin{aligned} u_\alpha &= -6z^{11} + 14z^{10} - 15z^9 + 21z^8 - 23z^7 + 28z^6 - 25z^5 + 27z^4 - 20z^3 + 18z^2 - 13z + 10, \\ u_\beta &= -6z^{15} + 41z^{14} - 125z^{13} + 266z^{12} - 474z^{11} + 711z^{10} - 935z^9 + 1088z^8 \\ &\quad - 1127z^7 + 1043z^6 - 857z^5 + 620z^4 - 389z^3 + 202z^2 - 85z + 21, \end{aligned}$$

the identities  $s^2 - (1 + 4u^2) = c \cdot f$  hold in  $\mathbb{Z}[z]$  with explicit cofactors  $c$ ; for  $\alpha$ ,

$$c_\alpha = -48z^9 + 20z^8 + 12z^7 + 104z^6 + 12z^5 + 104z^4 + 12z^3 + 56z^2 - 16z + 40.$$

Each identity is a single polynomial multiplication. All four are verified in Lean (§6).  $\square$

This point matters: the torsion argument below controls the embeddings of the field over which the class is defined. Because  $s_\alpha, s_\beta$  lie in  $F$  itself, the conclusion transports under *every* embedding simultaneously, with no place-by-place analysis.

## 4 The two classes are torsion

This section contains the new content. Let  $D$  denote the Bloch–Wigner dilogarithm, so that the Borel regulator of the extended Bloch element  $\xi = \sum_j [\widehat{z}_j]$  at an embedding  $\sigma$  is  $D_\sigma(\xi) = \sum_j D(\sigma(z_j))$ . Recall

$$D(\bar{z}) = -D(z), \quad D\left(\frac{1}{1-z}\right) = D(z), \quad D(z) = 0 \text{ for } z \in \mathbb{R}. \quad (7)$$

**Lemma 4** (Unit circle). *Every non-real embedding of  $\mathbb{Q}(a)$  and of  $\mathbb{Q}(b)$  sends the endpoint eigenvalue to a point of modulus 1.*

*Proof.*  $f_\alpha$  and  $f_\beta$  are palindromic, so each factors as  $f(z) = z^{n/2}g(z + z^{-1})$  for a polynomial  $g$  of degree  $n/2$  over  $\mathbb{Q}$ . For (1),

$$g_\alpha(w) = w^6 - 3w^5 - 2w^4 + 10w^3 - w^2 - 7w + 1,$$

and for (2),

$$g_\beta(w) = w^8 - 7w^7 + 14w^6 + w^5 - 25w^4 + 9w^3 + 12w^2 - 3w - 1.$$

Both are *totally real*:  $g_\alpha$  has 6 real roots and  $g_\beta$  has 8. A root  $z$  of  $f$  lies on  $|z| = 1$  if and only if  $z + z^{-1} \in [-2, 2]$ . Now  $g_\alpha$  has exactly 5 roots in  $[-2, 2]$  and one outside, giving 10 roots of  $f_\alpha$  on the unit circle and 2 real ones;  $g_\beta$  has exactly 6 in  $[-2, 2]$  and 2 outside, giving 12 on the circle and 4 real. These counts exhaust the degrees 12 and 16.  $\square$

**Lemma 5** ( $u$  is real). *In either chart,  $u(1/a) = u(a)$  identically. Consequently, at an embedding with  $|a| = 1$  one has  $\bar{u} = u$ , i.e.  $u \in \mathbb{R}$ .*

*Proof.* For the  $\alpha$  chart ( $L = a$ ,  $X = a^4$ ), (6) gives

$$u(a) = \frac{a + a^{12}}{a^4(a + a^4)} = \frac{1 + a^{11}}{a^4(1 + a^3)},$$

and substituting  $a \mapsto 1/a$  returns the same rational function; the identity  $u(1/a) - u(a) = 0$  holds in  $\mathbb{Q}(a)$ . For the  $\beta$  chart ( $L = b$ ,  $X = b^2$ ) one gets  $u(b) = b^{-2}(b^4 - b^3 + b^2 - b + 1)$ , again invariant under  $b \mapsto 1/b$ . On  $|a| = 1$  we have  $\bar{a} = 1/a$ , hence  $\bar{u} = u(\bar{a}) = u(1/a) = u(a) = u$ .  $\square$

**Lemma 6** ( $T$  and  $U$  are real). *At an embedding with  $|a| = 1$ ,  $T$  and  $U$  are real; hence  $D(T) = D(U) = 0$ .*

*Proof.*  $U = u$  is real by Lemma 5. Then  $1 + 4u^2 > 0$ , so  $\sqrt{1 + 4u^2}$  is real,  $r$  is real, and  $T = 1 - r^2$  is real. Apply (7).  $\square$

**Lemma 7** ( $V$  and  $W$  cancel). *At an embedding with  $|a| = 1$ ,  $W = 1/(1 - \bar{V})$ , and hence  $D(V) + D(W) = 0$ .*

*Proof.*  $|a| = 1$  gives  $|X| = 1$ , so  $\bar{X} = 1/X$ . With  $u$  real,

$$\bar{V} = \overline{\left(\frac{u}{X}\right)} = \frac{u}{\bar{X}} = uX, \quad \text{hence} \quad W = \frac{1}{1 - uX} = \frac{1}{1 - \bar{V}}.$$

By (7),  $D(W) = D(1/(1 - \bar{V})) = D(\bar{V}) = -D(V)$ .  $\square$

**Lemma 8** (The filling term). *Both endpoints satisfy the same  $(-1)$ -filling condition, and the flattenings are chosen compatibly along the arc, so that the extended Bloch classes  $\xi_\alpha, \xi_\beta$  are the classes of the filled representations. With that choice Neumann's surgery correction is already absorbed into the flattening data, and*

$$\text{GV}(\rho_\alpha) - \text{GV}(\rho_\beta) = 4 \text{Re}[\widehat{\mathcal{R}}(\xi_\alpha) - \widehat{\mathcal{R}}(\xi_\beta)]$$

*with no additional core term.*

*Remark 9.* It is worth being explicit about why this is stated as a choice of flattening and not as a vanishing computation. Neumann's core correction is *linear* in the core complex length  $\lambda_{\text{core}}$ , not quadratic: the contribution to the regulator is  $\frac{1}{2}\lambda_{\text{core}}\pi i \pmod{\pi^2\mathbb{Z}}$ . For  $\lambda_{\text{core}} = i\theta$  purely imaginary this is  $-\frac{1}{2}\theta\pi$ , which is real and *not* in general a rational multiple of  $\pi^2$ ; integral surgery does not force  $\theta/\pi \in \mathbb{Q}$ . So the earlier reasoning “ $\lambda_{\text{core}}^2$  is real, hence no volume contribution” is not available. The correct statement is the one above: the correction is the comparison between the *cusped* class and the *filled* class, and it does not reappear once one works with the filled class throughout. The 301-digit rationality of (8) is a consistency check on this: had an irrational term  $-\frac{\pi}{2}(\theta_\alpha - \theta_\beta)$  been silently omitted, the computed value could not have reconstructed to a rational.

**Lemma 10** (Orientation signs). *For the  $7_2$  triangulation used here,  $\varepsilon_j = +1$  for all four tetrahedra. In particular  $\varepsilon_V = \varepsilon_W$ , and the regulator is the unsigned sum  $\sum_j D(\sigma(z_j))$ .*

*Proof.* The sign  $\varepsilon_j$  is the orientation of tetrahedron  $j$  relative to the orientation of the manifold. It is combinatorial data fixed by the oriented triangulation, and it does *not* change along the deformation arc — in particular it must not be read off the sign of  $\text{Im } z_j$ , which degenerates exactly where the tetrahedra become flat, which is where our endpoints sit.

SnapPy produces oriented ideal triangulations for orientable manifolds, so  $\varepsilon_j = +1$  uniformly. This is confirmed by the volume identity  $\text{Vol}(M) = \sum_j \varepsilon_j D(z_j)$  at the complete structure: the *all-plus* sum reproduces the volume,

$$\sum_{j=1}^4 D(z_j) = 3.331744231641\dots, \quad \text{Vol}(7_2) = 3.331744231600\dots,$$

agreeing to  $4 \times 10^{-11}$ , and likewise for the  $(-1/2)$ -filling,  $\sum_j D(z_j) = 2.582170632117\dots$  against  $\text{Vol}(S_{-1/2}^3(7_2)) = 2.582170632120\dots$ , agreeing to  $3 \times 10^{-12}$ . A single sign flip would change the all-plus sum by  $2D(z_j) \geq 0.79$ , so no  $\varepsilon_j$  can be  $-1$ . See `scripts/orient_signs.py`.  $\square$

*Remark 11.* This step is worth isolating because it is the one place where the shape identity  $W = 1/(1 - \bar{V})$  is *not* enough. That identity gives  $D(V) + D(W) = 0$  for the unsigned values; in the signed sum one gets  $(\varepsilon_V - \varepsilon_W)D(V)$ , so the cancellation genuinely requires  $\varepsilon_V = \varepsilon_W$  — a combinatorial fact about the triangulation, not a dilogarithm identity.

**Theorem 12** (Torsion). *The extended Bloch classes  $\xi_\alpha$  and  $\xi_\beta$  of the two endpoint representations are torsion. Consequently*

$$\operatorname{Re}[\Delta\mathcal{R}] \in \pi^2\mathbb{Q}.$$

*Proof.* Let  $\sigma$  be an embedding of the endpoint field. If  $\sigma$  is real, all four shapes are real and  $D_\sigma = 0$  by (7). If  $\sigma$  is non-real, then  $|\sigma(a)| = 1$  by Lemma 4, so by Lemmas 6–10,

$$D_\sigma(\xi) = D(T) + D(U) + D(V) + D(W) = 0 + 0 + D(V) - D(V) = 0,$$

and by Lemma 8 the passage from the relative class to the closed class adds nothing to the Bloch–Wigner value. Thus the Borel regulator of  $\xi$  vanishes at every complex embedding, and a Bloch element whose Borel regulator vanishes at all complex embeddings is torsion. The Rogers regulator of a torsion class lies in  $\pi^2\mathbb{Q}$ .  $\square$

*Remark 13.* Numerically,  $|D_\sigma| < 10^{-60}$  at each of the five conjugate pairs of  $\mathbb{Q}(a)$ ; Theorem 12 explains this as an identity rather than a coincidence. We also checked that no embedding of  $\mathbb{Q}(a)$  gives  $D_\sigma = \pm \operatorname{Vol}(S_{-1/2}^3(7_2)) = \pm 2.58217063\dots$ , i.e. the  $\alpha$  component does not carry the discrete faithful character. (The image of  $\rho_\alpha$  is in fact *non-discrete*: the trace of a commutator word is an algebraic integer of degree 6 that is not of the form  $\zeta + \zeta^{-1}$  for any root of unity, so an elliptic element has infinite order. Non-discreteness does not obstruct torsion of the Bloch class.)

## 5 Denominator bound and conclusion

**Proposition 14** (Denominator). *Let  $F_\alpha = \mathbb{Q}(a)$  and  $F_\beta = \mathbb{Q}(b)$ . Then*

$$w_2(F_\alpha) = 120 = 2 \cdot 2^2 \cdot 3 \cdot 5, \quad w_2(F_\beta) = 408 = 2 \cdot 2^2 \cdot 3 \cdot 17,$$

*with  $\operatorname{disc} F_\alpha = -5^6 \cdot 59 \cdot 3881^2$  and  $\operatorname{disc} F_\beta = 17^{14} \cdot 1597$ . By the identification of the extended Bloch group with  $K_3^{\operatorname{ind}}$  and the Merkurjev–Suslin torsion formula  $|K_3^{\operatorname{ind}}(F)_{\operatorname{tors}}| = w_2(F)$ , the order of each class divides the corresponding  $w_2$ , so the denominator of  $\operatorname{Re}[\Delta\mathcal{R}]/\pi^2$  divides*

$$Q = \operatorname{lcm}(120, 408) = 2040.$$

**Theorem 15** (Main).  $\int_\alpha^\beta \left( \log x \frac{dy}{y} - \log y \frac{dx}{x} \right) = \frac{4\pi^2}{85}.$

*Proof.* By Theorem 12 and Proposition 14,  $\operatorname{Re}[\Delta\mathcal{R}]/\pi^2$  is a rational number with denominator dividing  $Q = 2040$ . Two distinct rationals with denominator at most  $Q$  differ by at least  $1/Q^2$ , so such a rational is determined uniquely by any approximation of error less than  $1/(2Q^2) = 1.20 \times 10^{-7}$ .

Evaluating (5) at 1000 bits of working precision from the exact endpoint data gives

$$\frac{\operatorname{Re}[\Delta\mathcal{R}]}{\pi^2} = -0.0470588235294117647058823529411764705882352941176470588235294\dots$$

with  $|\operatorname{Re}[\Delta\mathcal{R}]/\pi^2 + \frac{4}{85}| = 1.63 \times 10^{-301}$ , far below the separation threshold. Rational reconstruction with denominator bound  $Q$  therefore returns *uniquely*

$$\frac{\operatorname{Re}[\Delta\mathcal{R}]}{\pi^2} = -\frac{4}{85}. \quad (8)$$

(Consistency:  $85 \mid 2040$ , with quotient 24.) By (5),  $I = -\Delta\mathcal{R} = 4\pi^2/85$ .  $\square$

**Corollary 16.**  $\operatorname{GV}(\rho_\alpha) = \frac{74\pi^2}{15}$ , by (4).

*Remark 17* (Robustness of the denominator bound). The passage from “ $\xi$  is torsion of order dividing  $m$ ” to “the Rogers value lies in  $\frac{1}{m}\pi^2\mathbb{Z}$ ” is normalisation-sensitive: with Neumann’s  $\widehat{\mathcal{R}}$ , whose period lattice is  $\pi^2\mathbb{Z}$ , the denominator divides  $m = w_2(F)$  exactly as used above, but other normalisations of the Cheeger–Chern–Simons class carry conventional factors of 6, 12 or 24.

None of this affects the conclusion. The numerical certificate has error  $1.63 \times 10^{-301}$  while the separation of rationals of denominator at most  $Q$  is  $1/Q^2$ , so (8) follows from *any* denominator bound  $85 \leq Q \leq 10^{150}$  (the sharp ceiling is  $\sqrt{1/(2\varepsilon)} = 1.75 \times 10^{150}$ ) — roughly 146 orders of magnitude of slack. A stray factor of 6 or 24, or indeed of  $10^{146}$ , changes nothing. The Lean theorem `regulator_quotient_eq_robust` states exactly this, and is instantiated there at the inflated bound  $Q = 24 \cdot 2040$ .

## 6 Lean verification

The directory `lean/` is a standalone Lake project pinned to Lean v4.29.0 and Mathlib v4.29.0. It builds with `lake exe cache get` followed by `lake build` (Build completed successfully (3295 jobs)), contains no `sorry` and no `native_decide`, and every one of the 37 audited declarations depends only on `[propext, Classical.choice, Quot.sound]`.

|                        |                                                                  |
|------------------------|------------------------------------------------------------------|
| Lemma 4                | <code>TraceRoots.gAlpha_totally_real, gBeta_totally_real;</code> |
|                        | <code>UnitCircle.norm_eq_one_of_trace_real_abs_le_two</code>     |
| Lemma 5                | <code>ChartSymmetry.chartUAlpha_isReal_of_norm_one</code>        |
| Lemma 3                | <code>ShapeField.sPolyAlpha_sq, ShapeField.sPolyBeta_sq</code>   |
| Lemmas 6–7             | <code>ShapeCancellation.four_shape_sum_eq_zero</code>            |
| Lemmas 4–7 combined    | <code>MainTheorem.four_shape_sum_vanishes_of_trace_real</code>   |
| Theorem 15, final step | <code>MainTheorem.regulator_value</code>                         |
| Remark 17              | <code>regulator_quotient_eq_robust</code>                        |

Three inputs are stated as hypotheses rather than proved in Lean, and are cited above: the three Bloch–Wigner functional equations (7) (bundled as the structure `BlochWignerLaws`); the Merkurjev–Suslin denominator bound of Proposition 14; and the 301-digit numerical evaluation. The first is standard; the second enters only through Remark 17, which needs no more than its existence; the third is reproduced by `scripts/`.

## 7 Reproducing the computations

All computations are in `scripts/`; each is standalone (Sage, or Python with `mpmath/SnapPy`).

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|                                       |                                                     |
|---------------------------------------|-----------------------------------------------------|
| <code>falpha2.sage</code>             | derives (1) from $A(a^2, a)$                        |
| <code>confirm.sage</code>             | Lemma 4: unit-circle counts via $g_\alpha, g_\beta$ |
| <code>exact_proof.sage</code>         | Lemma 5: the identity $u(1/a) = u(a)$               |
| <code>mechanism.py</code>             | Lemmas 6–7 at every embedding                       |
| <code>filling_term.py</code>          | Lemma 8: core holonomy along the arc                |
| <code>shapefield.sage</code>          | Lemma 3: the square-root certificates               |
| <code>orient_signs.py</code>          | Lemma 10: all-plus sum reproduces the volume        |
| <code>alpha_component.py</code>       | no embedding realises $\text{Vol}(S^3_{-1/2}(7_2))$ |
| <code>disc_test.py</code>             | non-discreteness (cyclotomic trace test)            |
| <code>w2bound.sage</code>             | Proposition 14: $w_2 = 120,408$                     |
| <code>shapes300.sage, pin.sage</code> | the 301-digit evaluation and reconstruction         |

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## References

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